

Unified Theory Based on Time-Wave Dynamics (The P-model)

Part V: Future Observational Predictions

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1 Purpose of This Document

This chapter is an independent extension that fixes, for readers of the P-model, which future observations can decide acceptance or rejection next. It is built from the public canonical outputs of Parts II-IV. It does not repeat internal route logs, dated check records, or workflow tags. It records only the observables, required precision levels, and decision conditions.

The residual terms on the theory side remain canonical in Part III-A and Part III-B. Part V deals only with differences that have already been projected onto the observational side. Concretely, it fixes the future branches associated with the 4.8084% strong field shadow-diameter gap, the network gate for gravitational-wave polarization, the $\ln(1+z)$ term in cosmology, the direct $\kappa_a = 1/(2\pi)$ carry-over for independent galaxy rotation curves, the Born-rule deviation parameter $\chi = (m/m_{\text{crit}})^2$, the minimal connection in quantum information, the weak-field gravity-quantum differential table, and the reconnection conditions for distance-indicator premises.

For first-time readers: this chapter summarizes eight future observations that can decide whether the P-model survives. Each section is divided into **Why it matters**, **Technical specification**, and a reader-facing conceptual figure. Reading only the **Why it matters** blocks is enough to capture the overall map. The decision words used in this chapter mean the following:

- **Pass:** consistent with the currently available data
- **Watch:** not yet decided because the data are still insufficient
- **Reject:** inconsistent with data of sufficient precision and moving toward rejection
- **Reference:** not a direct decision target yet, but an anchor for future tests

2 Prediction Overview Table

This section gives the full map of the future-observation branches before the section-by-section details.

Observational domain	Frozen differential	Current state	Earliest decision window
Black-hole shadows	4.8084% shadow-diameter coefficient gap	Watch	2030-2031
Gravitational-wave polarization	simultaneous passage of the scalar-exclusion gate and the tensor-consistency gate	Watch	2027-2030
Cosmological $\ln(1+z)$ term	sign-stable residual across high- z tracers	Watch	2027-2035
Independent galaxy rotation curves	direct carry-over of $\kappa_a = 1/(2\pi)$	re-check current external samples	Pass on 2026-2032
Macroscopic quantum interference	direct Born-rule audit at $\chi \gtrsim 0.1$ -1	Reference	2035+
Quantum-computing experiments	gate-loss / depth limit implied by the minimal connection	Reference	2026-2032
Weak-field gravity-quantum observables	P-model-specific differentials in atom interferometers and optical lattice clocks	Watch	2026-2035+
Distance-indicator premise audit	decomposition of the DDR tension into lighter observational premises	Watch	2027-2030

For readability, a plain-language summary is separated into the next table.

Observational domain	One-line explanation
Black-hole shadows	Sharper black-hole images should reveal whether the P-model shadow is measurably larger than the GR shadow.
Gravitational-wave polarization	Measuring the vibration pattern of gravitational waves should test whether the network supports the P-model polarization gate.
Cosmological $\ln(1+z)$ term	High-redshift surveys should reveal the subtle residual that is unique to the P-model background law.
Independent galaxy rotation curves	A second galaxy sample should test whether rotation curves can still be explained without dark matter using the same frozen coefficient.
Macroscopic quantum interference	Heavy-molecule interference experiments should test whether the Born rule begins to fail at large mass.

Continued on next page

Observational domain	One-line explanation
Quantum-computing experiments	As quantum hardware improves, it should become clear whether benchmark performance approaches or crosses the P-model lower bound.
Weak-field gravity-quantum observables	Ultra-precise clocks and atom interferometers should test P-model-specific weak-field quantum differentials.
Distance-indicator premise audit	Lower-premise probes should separate whether the distance-indicator tension comes from the measurements or from the model.

The overall ordering is summarized again in Figure 9.

3 Black-Hole Shadows (ngEHT / BHEX, 2030-2031)

Why it matters

The first direct black-hole shadow image was released in 2019. The P-model predicts that the shadow size is about 5 percent larger than the value expected from general relativity. Current instruments are still too noisy to separate the two cases, but the next-generation EHT and related photon-ring programs are expected to reach the required regime around 2030. If future observations confirm a shadow that is larger by about 5 percent, the result supports the P-model branch. If the future shadow diameter remains consistent with the GR prediction, this prediction moves toward rejection.

Technical specification

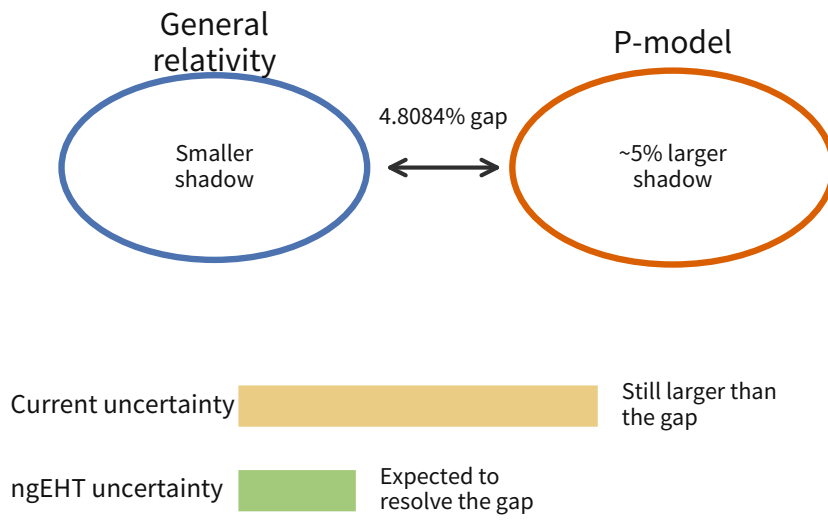
registry key: eht_shadow_compare / branch kind: **strong-field differential observable** / score kind: required_sigma_gate

The current EHT status is dominated by the ring-shadow correspondence systematic κ : the differential itself is already frozen, but the decision is not yet closed. The next decision surface is therefore a high-resolution photon-ring program such as ngEHT and BHEX.[1][2][3][4][5]

Item	Frozen content
P-model prediction	Adopt the shadow-diameter coefficient gap $ C_P - C_{\text{ref}} /C_{\text{ref}} = 4.8084\%$ as the combined coefficient difference established jointly by the Part II black-hole-shadow section and the joint strong-field higher-order-constraint section.
Current state	The dominant systematic is currently $\sigma(\kappa) \approx 0.12$, which is still larger than the differential itself.
Decision condition	A 3σ discrimination requires an overall 1σ uncertainty of roughly 1.6% or better; operationally this means compressing the systematic to about $\sigma(\kappa) \approx 0.011$.
Decision window	The shortest realistic window is 2030-2031, when photon-ring-resolving programs of the ngEHT / BHEX class become available.

Supplement: the target of this branch is not whether the image merely looks similar to GR, but whether the 4.8084% coefficient gap can be separated directly after systematic control.

Black-hole shadow differential



A direct shadow-diameter measurement is the decisive test.

Figure 1: Conceptual view of the black-hole shadow differential

4 Gravitational-Wave Polarization (LIGO / Virgo / KAGRA, 2027-2030)

Why it matters

Gravitational waves have already been detected, but their polarization content has not yet been measured sharply enough to close the network gate used by the P-model. The P-model predicts a definite polarization pattern that can only be tested cleanly when at least three detectors provide sufficiently good sky coverage. That makes the detector network after 2027 the key decision surface. If the required polarization pattern is confirmed by a three-detector network, the P-model survives this gate. If scalar modes cannot be excluded, or if tensor consistency fails, this branch moves toward rejection.

Technical specification

registry key: `gw_polarization_h1_l1_v1_network_audit` / branch kind: `network polarization gate` / score kind: `hard-gate boolean`

This branch is closed as a detector-network coverage problem rather than a new theory derivation. At the moment the coverage of highly correlated three-detector events is still too thin, so the most important next step is better event coverage in future runs.[7][6]

Item	Frozen content
P-model prediction	The acceptance condition established in Part II Section 4.11 is simultaneous passage of the scalar-exclusion gate and the tensor-consistency gate.
Current state	There are only two usable events at present; the maximum scalar-overlap proxy is 0.500 and H1/L1 gives $ r _{\max} = 0.749$, so both main gates remain unpassed and the branch stays at Watch.
Decision condition	Keep sufficient event coverage while driving the scalar-overlap proxy stably below 0.5, and pass both main gates simultaneously in a three-detector network.
Decision window	The first realistic decision window is 2027-2030, as the O5/O6-era three-detector network continues to expand.

Supplement: the outcome here is a survival condition for the P-model. Scalar exclusion and tensor consistency must both hold at the same time.

Gravitational-wave polarization gate

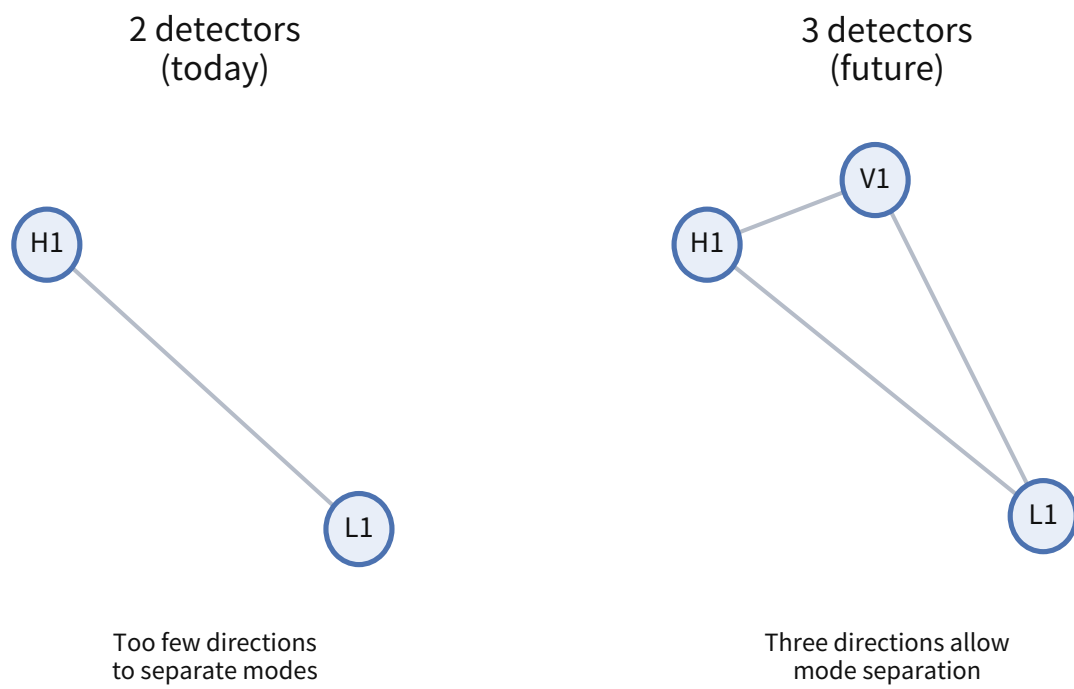


Figure 2: Conceptual view of the gravitational-wave polarization gate

5 The $\ln(1+z)$ Cosmology Term (DESI / Euclid / Roman, 2027-2035)

Why it matters

The P-model predicts a background-law pattern that is slightly different from standard Λ CDM expansion. The difference is expected to become visible only at high redshift, where the special $\ln(1+z)$ term matters. Large surveys such as DESI, Euclid, and Roman are therefore the natural places to test this branch. If multiple independent surveys recover the same high- z residual with stable sign, the result supports the P-model branch. If the residual does not appear in any such survey, this cosmological branch moves toward rejection.

Technical specification

registry key: `cosmology_desi_dr1_bao_promotion_check` / branch kind: `high-z background-law differential` / score kind: `promotion check`

The key issue is not low-redshift compressed-distance agreement. The relevant test is whether primary statistics containing high-redshift tracers recover the $\ln(1+z)$ differential in a tracer-stable way. DESI DR1 fixed the entry point, but it has not yet satisfied the promotion condition.[11][8][12]

Item	Frozen content
P-model prediction	Using the late-time background law from Part II Section 5.3, track a high-redshift residual in $\ln(1+z)$ that differs from GR / Λ CDM across multiple tracers.
Current state	DESI DR1 Stage B gives $z_{\text{score,combined}} \approx -11.13 \cdots + 0.70$ for LRG3+ELG1 and $z_{\text{score,combined}} \approx +0.68 \cdots + 1.35$ for Ly α QSO, leaving the number of passed tracers at 0/2 and the branch still at Watch.
Decision condition	Secure multiple high-redshift tracers with $ z \geq 3$ while keeping the sign stable across covariance-estimation choices.
Decision window	The observational window is 2027-2035, when continued DESI coverage, realized Euclid forecasts, and Roman high- z cross-checks become available together.

Supplement: the target is not a local significance excursion in one dataset, but a sign-stable differential residual across multiple tracers.

The $\ln(1 + z)$ cosmology term

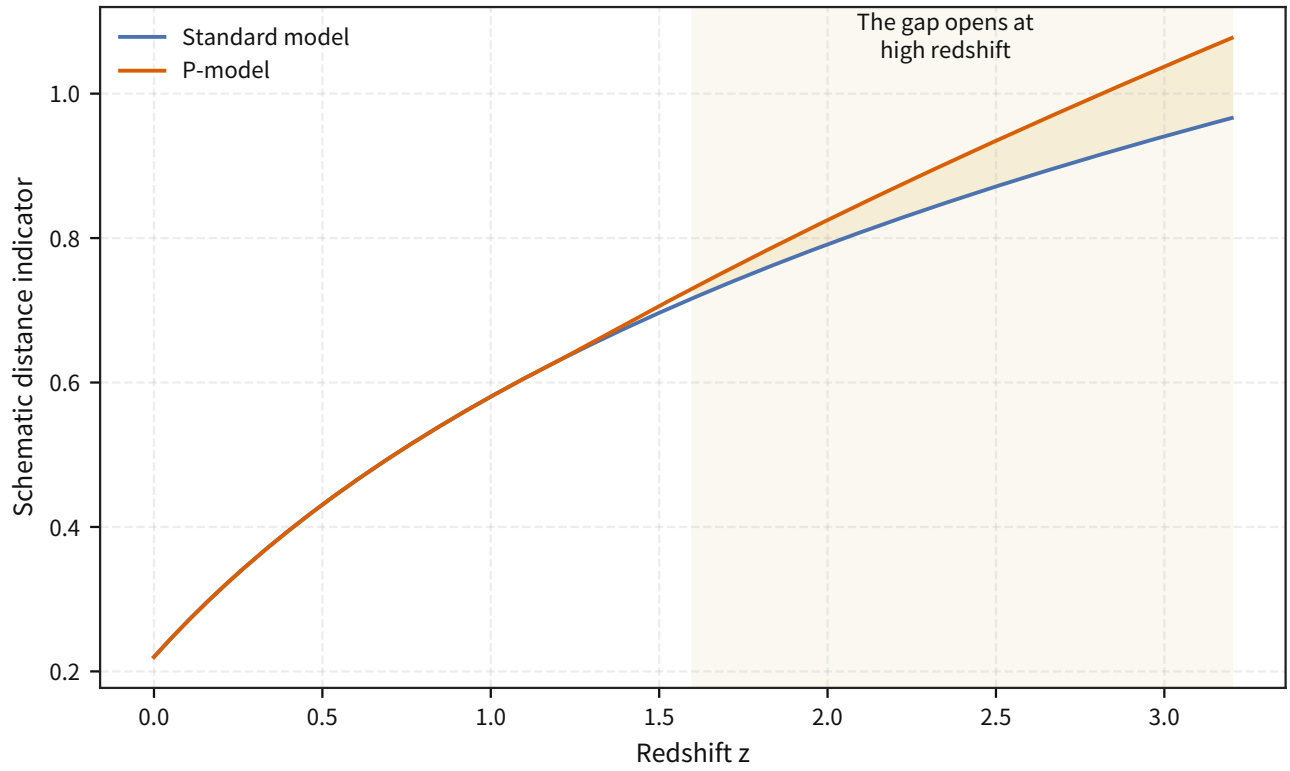


Figure 3: Conceptual view of the cosmological $\ln(1 + z)$ differential

6 Direct κ_a Verification with Independent Galaxy Rotation Curves (2026-2032)

Why it matters

Galaxies rotate too fast at large radius to be explained by luminous matter alone. The standard explanation introduces dark matter, but the P-model attempts to explain the same rotation curves using only the time-wave background law. This has already worked on the SPARC sample of 175 galaxies. The next test is whether the same frozen coefficient also survives on an independent sample. If the same coefficient explains another galaxy sample, the dark-matter-free branch is strengthened. If it fails outside SPARC, the branch moves toward rejection as a sample-specific accident.

Technical specification

registry key: `mass_origin_dark_matter_elimination_declaration_gate` / branch kind: `independent galaxy follow-through` / score kind: `direct coefficient carry-over`

The observational anchor of this branch is whether the frozen $\kappa_a = 1/(2\pi)$ established on SPARC can be carried over directly to an independent sample. The future test is not merely that SPARC worked, but that the same coefficient survives under the same baryonic interface on an external sample.[18]

Item	Frozen content
P-model prediction	Carry the frozen coefficient $\kappa_a = 1/(2\pi)$ from Part II Section 4.14 directly into an independent galaxy sample without refitting it.
Current state	On the SPARC set of 175 galaxies, 150/175 improve over baryons-only fits and the median χ^2/ν also improves, so the current branch is Pass.
Decision condition	Reproduce the improvement fraction and the global goodness metrics on a non-SPARC sample using the same baryonic preprocessing and the same frozen κ_a .
Decision window	The natural window is 2026-2032, when additional rotation-curve samples and compatible baryonic interfaces become available.

Supplement: this section does not directly claim the nonexistence of dark matter. It tracks whether the P-model coefficient can be carried over without retuning.

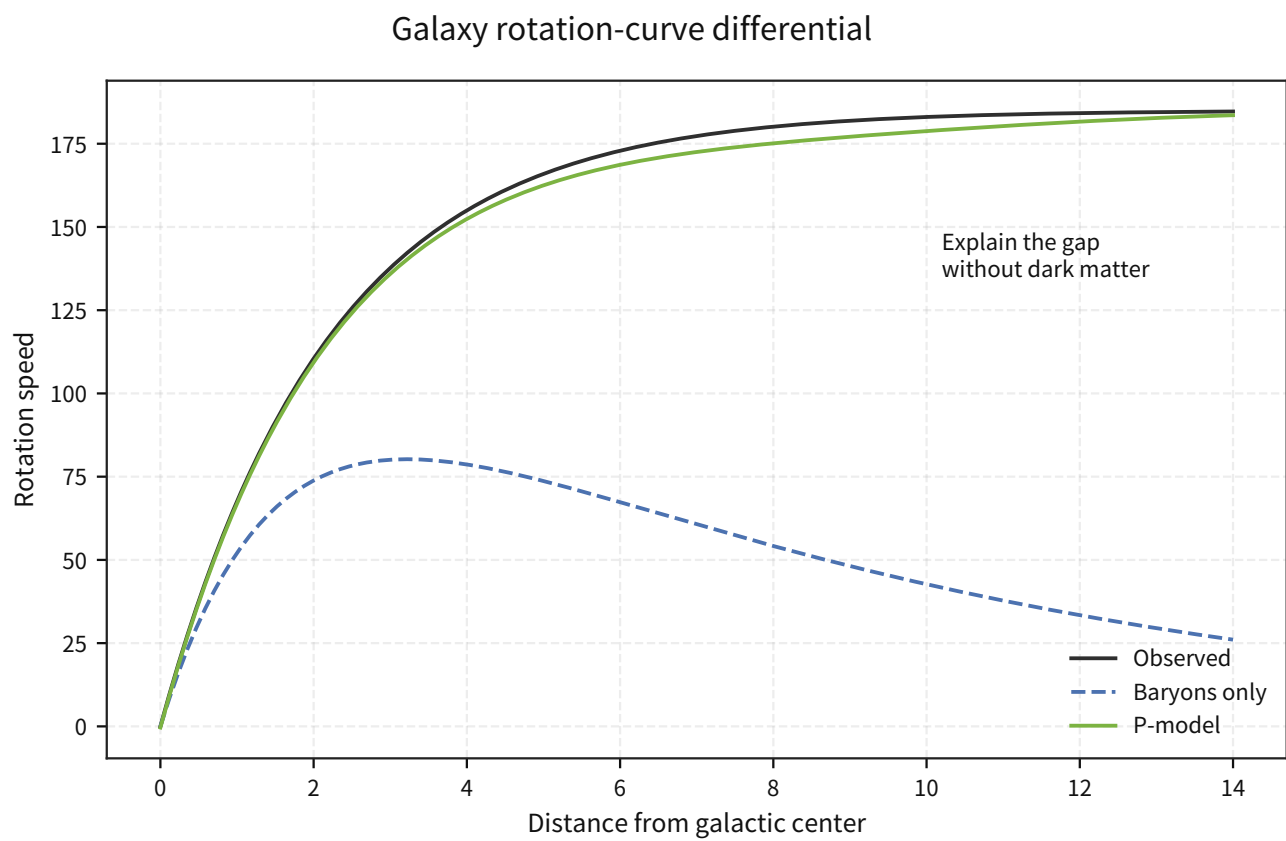


Figure 4: Conceptual view of the galaxy rotation-curve differential

7 Macroscopic Quantum Interference (MAQRO / OTIMA, 2035+)

Why it matters

Electrons and atoms behave like waves and produce interference fringes. The question for this branch is what happens when the mass scale becomes much larger. The P-model predicts that interference patterns should begin to degrade once a critical mass regime is crossed. Testing that regime requires long-duration, high-mass interference experiments, probably in space, so the realistic window is 2035 or later. If interference begins to fail for sufficiently heavy objects, the result supports the P-model branch. If clean interference survives throughout the predicted mass regime, this mass-threshold branch moves toward rejection.

Technical specification

registry key: mass_origin_gravitational_self_binding_solver_statement / branch kind: **Born-rule deviation threshold** / score kind: **chi threshold**

Direct Born-rule deviation is expected only after pushing the control parameter $\chi = (m/m_{\text{crit}})^2$ into the range 0.1-1. That is far beyond the current molecular-interference regime. OTIMA is therefore only a distant anchor point, while a MAQRO-class platform is closer to the actual decision surface.[16][17][19][20]

Item	Frozen content
P-model prediction	The control parameter fixed in Part III-B Section 5.2 is $\chi \equiv (E_G/\hbar)T = (m/m_{\text{crit}})^2$.
Current state	In the OTIMA-like regime, $m_{\text{crit}} = 7.6 \times 10^{11}$ amu and $\chi = 1.7 \times 10^{-8}$, far outside the detection window; even in the MAQRO-like regime, the representative values are $m_{\text{crit}} = 2.4 \times 10^{10}$ amu and $\chi = 0.17$.
Decision condition	Reach $\chi \gtrsim 0.1$ -1 while holding phase / visibility errors to $\lesssim \chi/3$, so the presence or absence of Born-rule deviation can be tested directly at 3σ .
Decision window	The first realistic window is 2035+, when space-based macroscopic interferometry becomes technologically plausible.

Supplement: the current null result from molecular interference is not a counterexample to the P-model. It only reflects the fact that the present experiments still operate at $\chi \ll 1$.

Macroscopic interference breakdown

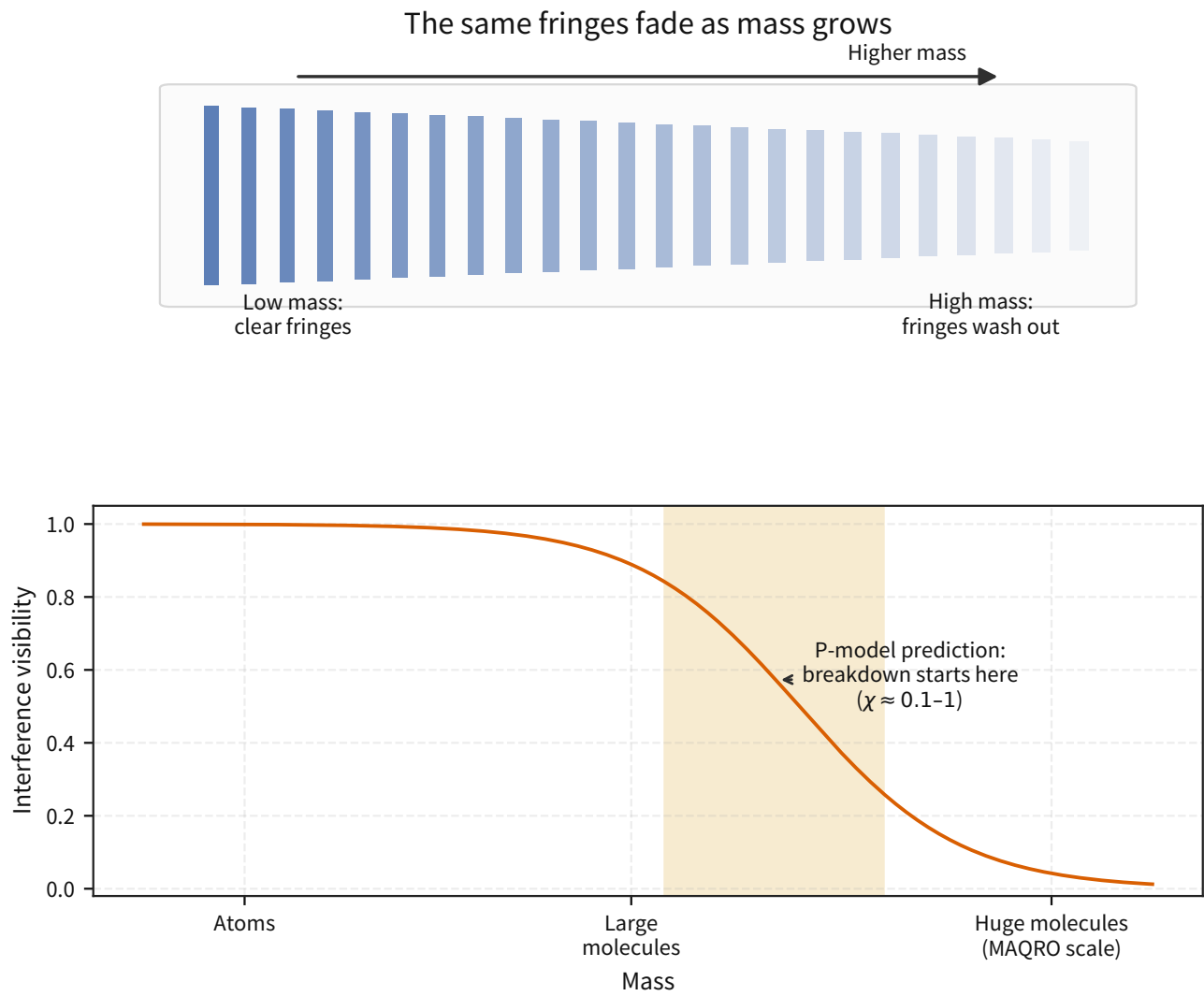


Figure 5: Conceptual view of the macroscopic-interference audit band

8 Decoherence Audit through Quantum-Computing Experiments (2026-2032)

Why it matters

Quantum computers rely on entanglement and phase coherence. The P-model does not predict that quantum computation is impossible, but it does predict a lower bound on phase scattering, namely on decoherence. As hardware performance improves, the question becomes whether public benchmarks eventually hit that floor. If benchmark progress saturates near the predicted floor, the result supports the P-model branch. If platforms stably beat the floor by a wide margin, this lower-bound branch moves toward rejection.

Technical specification

registry key: `quantum_information_minimal_connection` / branch kind: `platform-level decoherence differential` / score kind: `entry-fixed lower bound`

In this branch the P-model does not predict the impossibility of quantum computing. Instead, it provides platform-specific lower-bound surfaces for gate fidelity and effective circuit depth. The next observational question is whether stable public benchmarks cross those surfaces.[19][20]

Item	Frozen content
P-model prediction	Part III-A Section 2.5.5 fixes for transmons $1 - F_{2q}^{(\phi)} \leq 3.33 \times 10^{-4}$ together with $d_{\max} \geq 1.5 \times 10^3$; for trapped ions $1 - F_{2q}^{(\phi)} \leq 7.6 \times 10^{-8}$ together with $d_{\max} \geq 6.58 \times 10^6$; and for the photonic branch $T_2 = 5.003 \times 10^{-4}$ s together with a pair-distance budget of 37.5 km.
Current state	These values are fixed as entry conditions on the theory side, but they are still weak as one-to-one public benchmark metrics.
Decision condition	Bring platform benchmarks into a common observable basis for gate loss, phase scattering, and circuit depth, so that they can be compared against the lower-bound surfaces at 3σ .
Decision window	The first practical audit window is 2026-2032, as transmon, trapped-ion, and photonic benchmark standards mature.

Supplement: the point of this branch is not whether quantum computing exists, but whether platform-specific decoherence behavior crosses the P-model lower bound.

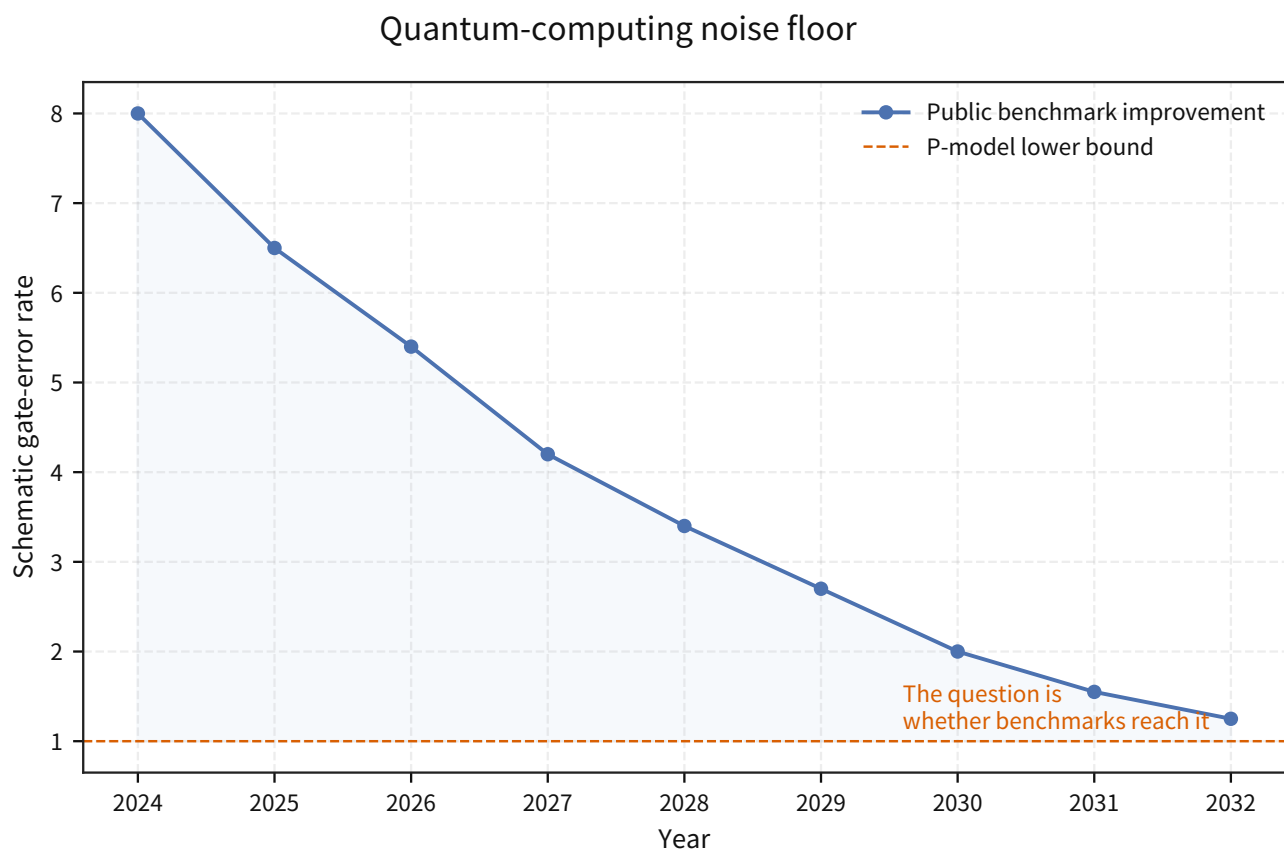


Figure 6: Conceptual view of the quantum-computing lower bound

9 Weak-Field Gravity-Quantum Differentials (Atom Interferometers / Optical Lattice Clocks, 2026-2035+)

Why it matters

Ultra-precise atom interferometers and optical lattice clocks can probe the boundary where gravity and quantum physics meet. The P-model predicts, in that boundary regime, a specific differential that is not present in the standard picture, together with the observing geometry that should expose it most clearly. Current instruments are not yet precise enough, but the branch is expected to enter the observable range in the 2030s. If the P-model-specific geometry reveals the predicted shift, the result supports the branch. If sufficiently precise measurements still fail to find the shift, this weak-field differential branch moves toward rejection.

Technical specification

registry key: gravity_quantum_differential_prediction_table / branch kind: **weak-field quantum differential** / score kind: **row-specific precision gate**

This branch is not about re-explaining the known COW phase or standard leveling experiments. It asks how far the P-model-specific differentials stand from current precision levels. Atom interferometers and optical lattice clocks therefore serve as row-wise future ranking gates.[14][15]

Item	Frozen content
P-model prediction	In the Part III-B differential table, the required precision for the P-model-specific local arm in atom interferometry is 3.92×10^{-12} in fractional phase; for optical lattice clocks the main 399 m row gives $\delta(\Delta f/f) = -6.05 \times 10^{-23}$, while the solar lever-arm row requires an absolute precision of 1.30×10^{-14} .
Current state	The Earth-leveling 399 m line remains far from the required precision, while the solar lever-arm line enters the range of current clock precision if the observing configuration can be realized.
Decision condition	Meet the required precision within 3σ on the P-model-specific rows themselves, not only on proxy rows that are already known in the standard picture.
Decision window	The observational window is 2026-2035+, as short-baseline atom interferometers and long-baseline optical lattice clock networks continue to improve.

Supplement: the priority is no longer to improve the main 399 m geometry alone. It is to realize observing geometries that sit closer to the P-model-specific rows.

Weak-field gravity-quantum gap

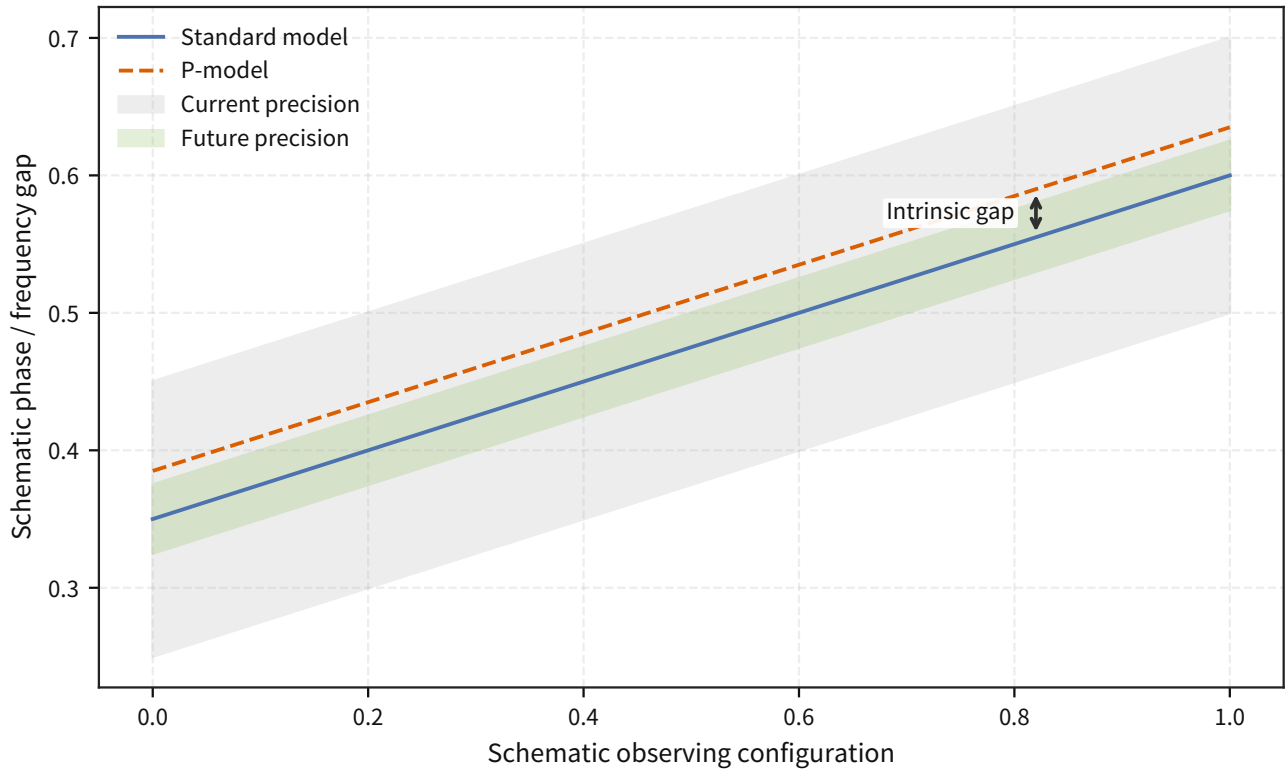


Figure 7: Conceptual view of the weak-field gravity-quantum gap

10 Distance-Indicator Premise Audit (Standard Sirens / JWST, 2027-2030)

Why it matters

Cosmic distances are often inferred from standard candles and related calibrated probes, yet the P-model prediction and the current data show a serious mismatch. At present it is still unclear whether that mismatch comes from the P-model itself or from the heavy premise stack built into the distance ladder. Lower-premise probes, especially standard sirens from gravitational waves together with JWST follow-up, are the key tools for separating those two possibilities. If the mismatch survives even after the premise stack is reduced, the cause is more likely on the P-model side and the branch moves closer to rejection. If the mismatch weakens once the premises are reduced, the problem may lie on the measurement side instead, and this branch remains open for the P-model.

Technical specification

registry key: `cosmology_distance_duality_constraints` / branch kind: `premise-sensitive reopen` / score kind: `premise-light cross-check`

The raw tension against the minimal P-model is already heavy. Part V does not downgrade that fact. Instead, it separates premise-heavy distance-ladder ingredients from lighter probes and assigns a reconnection condition for the next decisive test.[8][9][10][13]

Item	Frozen content
P-model prediction	Part II Section 5.3 fixes, as the representative case, a DDR tension reaching $z \approx -25.10$, which is a severe challenge to the minimal P-model.
Current state	The tension is large, but it still mixes multiple premises at once, including standard candles, standard rulers, calibration evolution, and opacity; the next step is therefore a lighter cross-check.
Decision condition	Use the geometric distance from standard sirens together with high-redshift age / spectroscopy cross-checks including JWST, so the dominant source of the DDR tension can be separated into a premise-side or model-side origin.
Decision window	The first reconnection window is 2027-2030, when multi-messenger standard sirens and JWST follow-up should both be available in larger numbers.

Supplement: this section does not claim that the current tension is mild. It decomposes a heavy tension into its premise layers so the next decisive test can target the correct source.

Premise-sensitive distance indicators

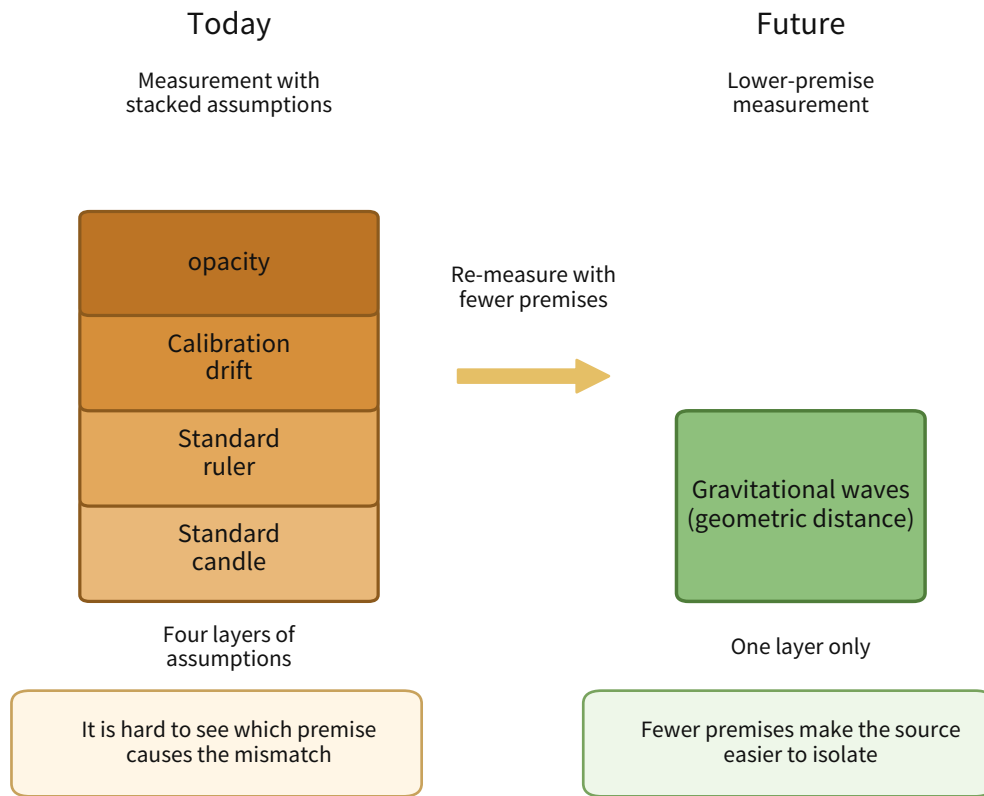


Figure 8: Conceptual view of the distance-indicator premise audit

11 Timeline Figure

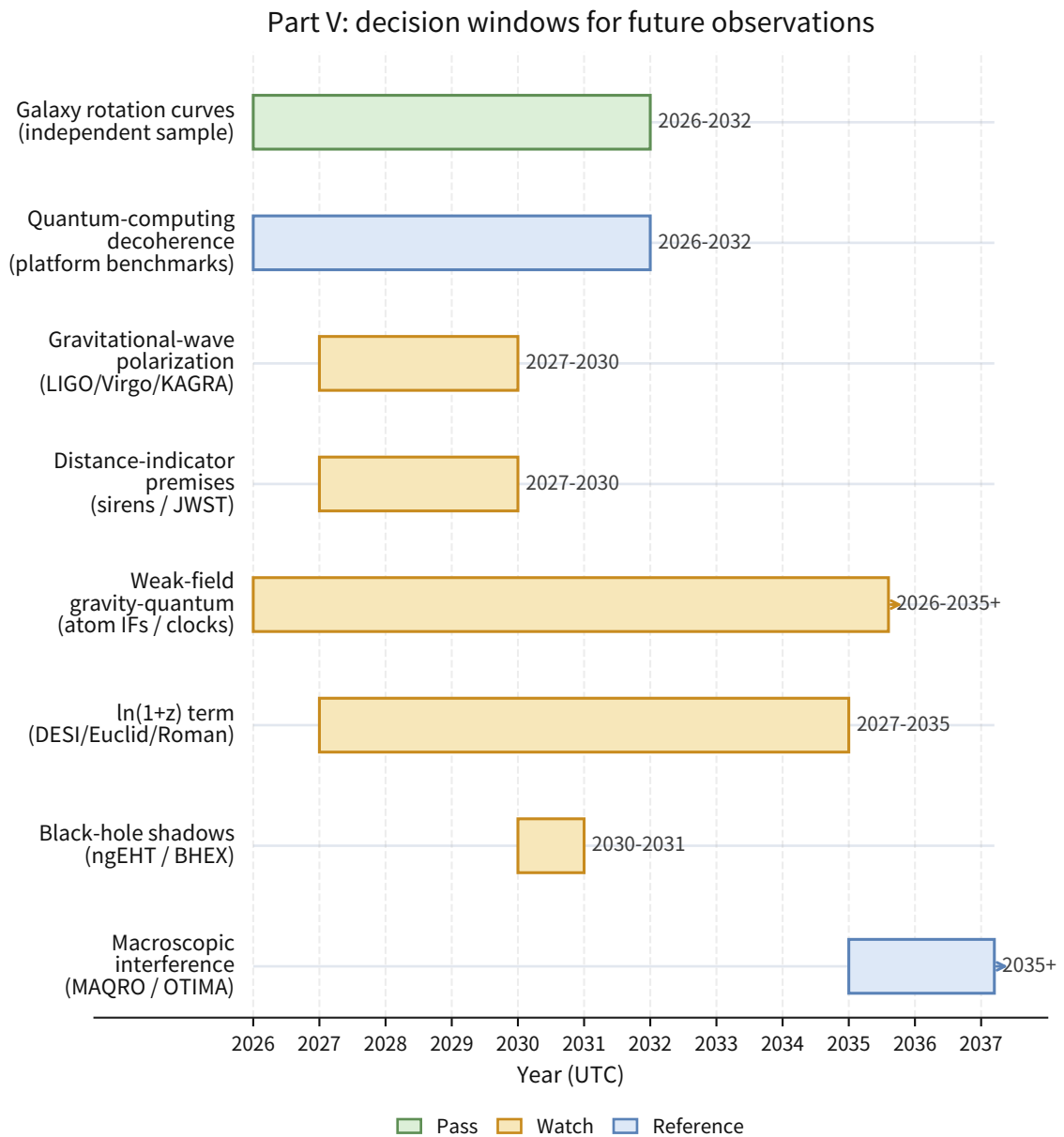


Figure 9: Timeline of the future observational decision windows in Part V

This figure is the reader-facing roadmap for the future predictions in Part V. It orders the eight observational branches by the period in which a decision is most likely to emerge. The colors encode the current state of each branch: green for Pass, yellow for Watch, and gray for Reference.

12 Update Rules (Part V)

1. Keep Part V as a reader-facing observational roadmap; do not insert internal route logs, dated check records, or *.number – .number* tag strings into the manuscript body.
2. Every prediction section must state **registry key /branch kind /score kind**, and must include the four required rows **P-model prediction /current state /decision condition /decision window**.
3. Do not invent new numbers in Part V; quote only frozen values that are already consistent with the public canonical outputs of Parts II-IV.
4. Keep the decision vocabulary fixed to Pass / Watch / Reject / Reference; do not introduce Part V-specific decision words.
5. Keep detailed theory-side residuals canonical in Part III-A and Part III-B; Part V records only observable differentials and reconnection conditions.
6. Keep the timeline PDF and the per-section conceptual PDFs as the canonical figure set for Part V; describe them as a roadmap or conceptual figures rather than as comparative result figures.

13 Data / Code Availability

The canonical source for the frozen values quoted in this chapter is the public artifact set of Parts II-IV together with the reference section of this paper. For primary-data acquisition routes, reproduction commands, verification summaries, and falsification-condition packs, see Part IV and the related operations documents.

The Part V-specific canonical figures and metrics JSON are the timeline artifacts corresponding to Figure 9 and the conceptual-figure artifact set listed in Sections 3-10.

References

- [1] Event Horizon Telescope Collaboration. “The persistent shadow of the supermassive black hole of M87. I. Observations, calibration, imaging, and analysis.” *Astronomy & Astrophysics* **681**, A79 (2024). DOI: 10.1051/0004-6361/202347932.
- [2] Event Horizon Telescope Collaboration. “The persistent shadow of the supermassive black hole of M87. II. Model comparisons and theoretical interpretations.” *Astronomy & Astrophysics* **693**, A265 (2025). DOI: 10.1051/0004-6361/202451296.
- [3] Johnson, M. D., et al. “Universal interferometric signatures of a black hole’s photon ring.” *Science Advances* **6**(12), eaaz1310 (2020). DOI: 10.1126/sciadv.aaz1310. arXiv:1907.04329.
- [4] Doeleman, S. S., et al. “Reference Array and Design Consideration for the Next-Generation Event Horizon Telescope.” *Galaxies* **11**(5), 107 (2023). DOI: 10.3390/galaxies11050107. arXiv:2306.08787. Accessed: 2026-04-05.
- [5] Lupsasca, A., Cárdenas-Avendaño, A., Palumbo, D. C. M., Johnson, M. D., Gralla, S. E., Marrone, D. P., Galison, P., Tiede, P., and Keeble, L. “The Black Hole Explorer: Photon Ring Science, Detection and Shape Measurement.” In: *Space Telescopes and Instrumentation 2024: Optical, Infrared, and Millimeter Wave* (Proc. SPIE **13092**, 130926Q, 2024). DOI: 10.1117/12.3019437. arXiv:2406.09498. Accessed: 2026-04-05.
- [6] Abbott, B. P., et al. “Observation of Gravitational Waves from a Binary Black Hole Merger.” *Physical Review Letters* **116**, 061102 (2016). DOI: 10.1103/PhysRevLett.116.061102. Accessed: 2026-01-06.
- [7] Gravitational Wave Open Science Center (GWOSC). Event API / strain data. Web: <https://gwosc.org/eventapi/>. Accessed: 2026-01-06.
- [8] Martinelli, M., et al. “Euclid: Forecast constraints on the cosmic distance duality relation with complementary external probes.” *Astronomy & Astrophysics* (2021). DOI: 10.1051/0004-6361/202039637. arXiv:2007.16153. Accessed: 2026-01-08.
- [9] Abbott, B. P., et al. (LIGO Scientific Collaboration and Virgo Collaboration). “GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral.” *Physical Review Letters* **119**, 161101 (2017). DOI: 10.1103/PhysRevLett.119.161101. arXiv:1710.05832. Accessed: 2026-01-10.
- [10] Hjorth, J., et al. “The distance to NGC 4993: the host galaxy of the gravitational-wave event GW170817.” arXiv:1710.05856 (2017). Accessed: 2026-01-10.
- [11] DESI Collaboration, Aghamousa, A., et al. “The DESI Experiment Part I: Science, Targeting, and Survey Design.” arXiv:1611.00036 (2016). Accessed: 2026-04-05.
- [12] NASA Science. “Roman.” Web: <https://science.nasa.gov/mission/roman-space-telescope/>. Accessed: 2026-04-05.
- [13] Gardner, J. P., Mather, J. C., Clampin, M., et al. “The James Webb Space Telescope.” *Space Science Reviews* **123**, 485-606 (2006). DOI: 10.1007/s11214-006-8315-7. Accessed: 2026-04-05.
- [14] Colella, R., Overhauser, A. W., and Werner, S. A. “Observation of gravitationally induced quantum interference.” *Phys. Rev. Lett.* **34**, 1472 (1975). DOI: 10.1103/PhysRevLett.34.1472.

- [15] arXiv:2309.14953v3 “Long-distance chronometric leveling with a portable optical clock” . arXiv: <https://arxiv.org/abs/2309.14953>. Accessed: 2026-01-25.
- [16] Kaltenbaek, R., Hechenblaikner, G., Kiesel, N., Romero-Isart, O., Schwab, K. C., Johann, U., and Aspelmeyer, M. “Macroscopic quantum resonators (MAQRO).” *Experimental Astronomy* **34**(2), 123-164 (2012). DOI: 10.1007/s10686-012-9292-3. arXiv:1201.4756. Accessed: 2026-04-05.
- [17] Haslinger, P., Dörre, N., Geyer, P., Rodewald, J., Nimmrichter, S., and Arndt, M. “A universal matter-wave interferometer with optical ionization gratings in the time domain.” *Nature Physics* **9**, 144-148 (2013). DOI: 10.1038/nphys2542. Accessed: 2026-04-05.
- [18] Lelli, F., McGaugh, S. S., and Schombert, J. M. “SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves.” *The Astronomical Journal* **152**, 157 (2016). DOI: 10.3847/0004-6256/152/6/157. arXiv:1606.09251. Accessed: 2026-04-05.
- [19] Pikovski, I., Zych, M., Costa, F., and Brukner, C. “Universal decoherence due to gravitational time dilation.” *Nature Physics* **11**, 668-672 (2015). DOI: 10.1038/nphys3366. arXiv:1311.1095.
- [20] Diósi, L. “A universal master equation for the gravitational violation of quantum mechanics.” *Physics Letters A* **120**(8), 377–381 (1987). DOI: 10.1016/0375-9601(87)90681-5.

Final Note

I have often spent time thinking deeply about philosophy as a discipline.

When one tries to understand things in depth, one eventually arrives at science.

And yet I have felt that science, as we have known it, still contains something that resembles philosophy.

That point is α , and it is the singularity.

Cosmology and quantum physics are the same physical reality, and yet they are framed in different ways.

That is why I feel there is something philosophical here.

Human beings are incomplete, and so they prefer incomplete things.

That itself can be called a kind of philosophy.

In this world there are only three spatial dimensions that describe the positions of objects, and time.

Everything began with that simple question.

It is acceptable not to be complete. Science too is something done by human beings.

There is only one thing that matters.

“People live for other people, and are kept alive by other people.”

To understand this deeply is, I believe, the essence of philosophy and of science.

That understanding is also the only morality that can lead to peace.

There is no god and no Buddha in this world.

There is no rank and no class.

What exists is only the wave called time, changing form and continuing to be.

Rather than clinging to something else,

to be grateful for being alive now,

and to be grateful that we are sustained by the people around us,

is the only path that leads to happiness.

Shunji Ogawa